

JEE MAIN REVISION MASTER: ELASTICITY

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Elasticity & Hooke's Law	9 PYQs	Solids
Standard	Fluid Statics & Dynamics (Bernoulli & Torricelli)	0 PYQs	Fluids
Standard	Surface Tension & Viscosity (Stokes' Law)	0 PYQs	Surface & Flow

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Elasticity & Hooke's Law	$Y = \frac{F/A}{\Delta L/L}$, $\Delta L = \frac{FL}{AY}$, $u = \frac{1}{2}\sigma\epsilon$, $\epsilon = \frac{\sigma^2}{2Y}$, $U = \frac{1}{2}F\Delta L$
Fluid Statics & Dynamics (Bernoulli & Torricelli)	$A_1 v_1 = A_2 v_2$, $P + \frac{1}{2}\rho v^2 + \rho gh = \text{const}$, $v_{\text{efflux}} = \sqrt{2gh}$, $R = 2\sqrt{h(H-h)}$
Surface Tension & Viscosity (Stokes' Law)	$\Delta P_{\text{drop}} = \frac{2T}{R}$, $\Delta P_{\text{bubble}} = \frac{4T}{R}$, $h = \frac{2T\cos\theta}{\rho g r}$, $v_t = \frac{2r^2(\rho - \sigma)g}{9\eta}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Elasticity & Hooke's Law (High Yield — 9 PYQs)

Core Principle: Stress-strain relationships, Young's modulus, shear modulus, bulk modulus, and energy stored in stretched wires.

■ **Caution / JEE Pitfall:** Elongation due to self-weight is $\Delta L = (\rho g L^2 / 2Y) = (MgL / 2AY)$ (half of an equivalent end load Mg).

■ **Speed Trick:** When two wires are in series, equivalent Y is obtained by adding extensions: $\Delta L = \Delta L_1 + \Delta L_2$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "There are two wires of same material and same length while the diameter of second wire is two times the diameter of the first wire. Then the..." [Ans: D]
- [2025 Shift] Q#2 (MCQ): "Two wires A and B are made of same material. Their diameters are in the ratio of 1:2 and lengths are in the ratio of 1:3. If they are s..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "A thick metal wire of density ρ and length L is hung from a rigid support. The increase in length of the wire due to its own weight i..." [Ans: N/A]

3.2 Fluid Statics & Dynamics (Bernoulli & Torricelli) (Standard Yield — 0 PYQs)

Core Principle: Continuity equation, Bernoulli's energy conservation principle, Torricelli's efflux theorem, and hydraulic lift.

■ **Caution / JEE Pitfall:** Speed of efflux is $\sqrt{2gh}$ where h is depth from the free liquid surface, NOT height from the ground!

■ **Speed Trick:** Maximum horizontal range of water jet occurs when hole is at mid-depth $h = H/2$, with $R_{\text{(max)}} = H$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "There are two wires of same material and same length while the diameter of second wire is two times the diameter of the first wire. Then the..." [Ans: D]
- [2025 Shift] Q#2 (MCQ): "Two wires A and B are made of same material. Their diameters are in the ratio of 1:2 and lengths are in the ratio of 1:3. If they are s..." [Ans: N/A]

3.3 Surface Tension & Viscosity (Stokes' Law) (Standard Yield — 0 PYQs)

Core Principle: Surface energy, excess pressure in drops and bubbles, capillary rise formula, and terminal velocity of falling spheres.

■ **Caution / JEE Pitfall:** A soap bubble in air has TWO liquid-air interfaces, so excess pressure is $\Delta P = (4T / R)$, while a liquid drop has ONE ($\Delta P = (2T / R)$).

■ **Speed Trick:** Terminal velocity $v_t \propto r^2$. When n identical droplets coalesce: $R = n^{1/3}r$, and surface energy is released: $\Delta U = 4\pi T r^2(n - n^{2/3})$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "There are two wires of same material and same length while the diameter of second wire is two times the diameter of the first wire. Then the..." [Ans: D]
- [2025 Shift] Q#2 (MCQ): "Two wires A and B are made of same material. Their diameters are in the ratio of 1:2 and lengths are in the ratio of 1:3. If they are s..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

There are two wires of same material and same length while the diameter of second wire is two times the diameter of the first wire. Then the ratio of extensions produced in the wires by applying same load will be

- (A) 1:1
- (B) 1:2
- (C) 2:1
- (D) 4:1 [Correct]

Correct Answer: D

Solution Analysis: For a wire, the extension produced by a load is given by $\Delta L = (FL / AY)$ Here, F is the load applied, L is the length of the wire, A is the area of cross-section, Y is Young's modulus. Both wires have the same material, so Y is the same. Both have the same length, and the same load is applied. So, $\Delta L \propto (1 / A)$ Now, area of cross-section of a wire is $A = \frac{\pi d^2}{4}$ So area is proportional to the square of diameter: $A \propto d^2$ Given, $d_2 = 2d_1$ So, $A_2 = 4A_1$ Since extension is inversely proportional to area, $\frac{\Delta L_1}{\Delta L_2} = \frac{A_2}{A_1}$